Project data

- No need to remove any initial biases for all six sensors of both TPI and NovAtel units for this trajectory. I run both right now and they are giving good results.
- No need to skip any parts of this trajectory.
- The initial position is simply obtained from the NovAtel integrated solution that is provided in the "INSPVA" file under the NovAtel folder. If someone is starting the trajectory from first epoch then, he/she can initialize using the position from the first epoch and so on. Records from different files match based on the GPS time stamp.
- Like above. The initial attitude (and velocity) information are obtained from the "INSPVA" file.
- Only for the KVH IMU. The f.y and w.y measurements are logged in the reverse direction. This is from the manufacturer. So, what we do is simply invert those by just writing f.y = -f.y and w.y = -w.y in our code after we load those data vectors.
- Another important note for the integration later. The VTI IMU data is logged at 20 Hz rate but the KVH IMU data is logged at 200 Hz rate. For the mechanization only, they can run at the full given rate as we have been discussing lately. For the integration they need to down sample to the GPS data rate.
- IMPORTANT: Units are **m/s$^2$ for Accelerometers** and **rad/sec for the gyros**.
- It is recommended to start with the high end IMU (KVH1750) data to make sure that the mechanization algorithm is working then you can switch to the data set from the low end IMU (VTI). This way you can avoid any effects of large sensor errors in the low end IMU that might cause the solution to look awkward and you would know in advance how the solution would look like since it is too close to the reference solution.
- Two issues are critical to your implementation. First, the start time and end time of your data need to be the same for the different sensors you are using at the same time. Second, the data sets need to be at the same sampling rate.
    - For the first issue, the data logging often starts at different time seconds for the several sensors aboard the test bed. So, you need to find the right time at which data is available from all sensors involved in your work. Let's say the logging started at:

StartTime_Odo  (for the odometer)

StartTime_VTI  (for the VTI unit)

StartTime_KVH  (for the KVH unit or NovAtel integrated system)

Then your start time in your implementation must be equal to or greater than the largest of these. We practically use:

StartTime = max (StartTime_Odo, StartTime_VTI, StartTime_KVH)

The data end time could be found similarly (but the smallest number this time).

EndTime = min (EndTime_Odo, EndTime_VTI, EndTime_KVH)

- For the second issue, make sure that your data sets from all sensors are at the same data rate. You are given the odometer data at a rate of 1 Hz. So, you need to down sample the VTI IMU data from 20 Hz to 1 Hz as well. You also need to down sample the KVH data from 200 Hz to 1 Hz. When you do so, remember to keep the time index referring the to the right sample so the data from all the sensors is synchronized. Also, remember that when we down sample we average the corresponding samples from the original data sets (e.g. 20 samples from VTI data) into one sample in the new data set but for the time index you need to pick only one number that represents the right time for the new sample.